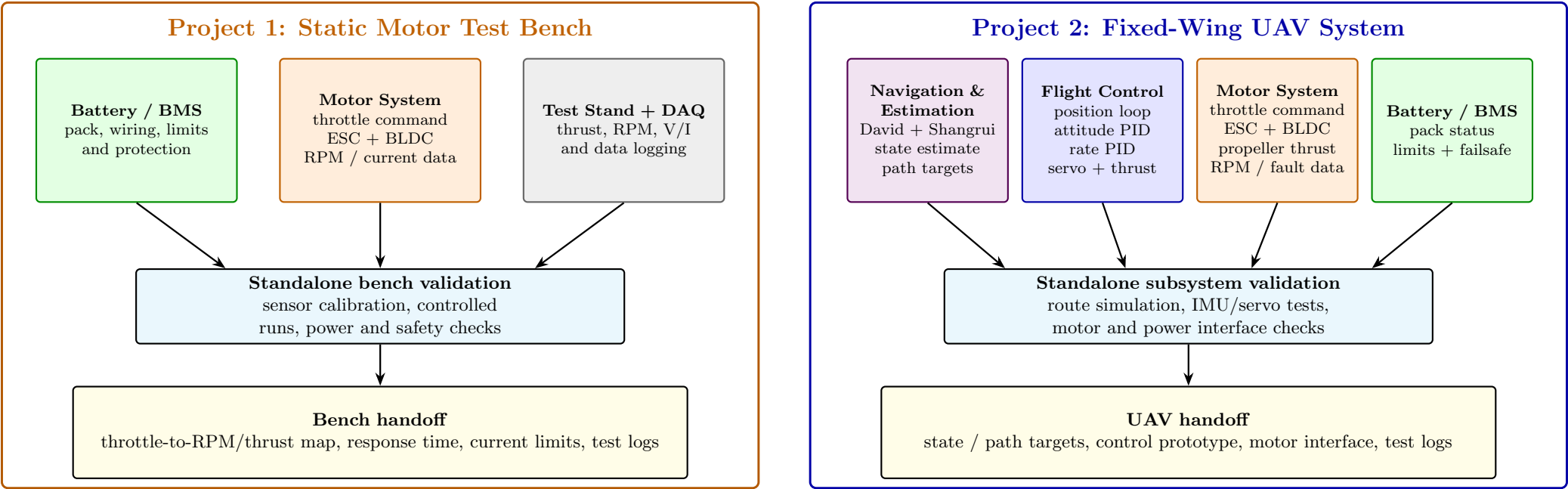
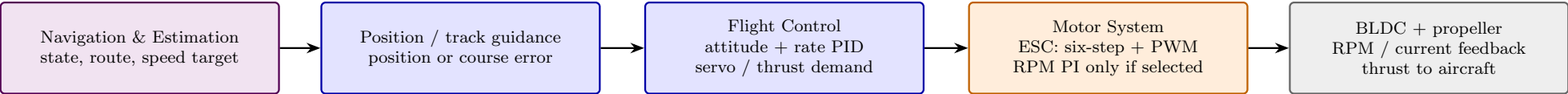


DARES Electronics: Two Projects Before Integration

The work packages below run in parallel. Each project validates its own subsystems and records a handoff package.



Control hierarchy and motor scope



Motor System: which loops to implement?

Function	Decision	Condition / measurement
Six-step + PWM	Current drive method; not a PID loop	Verify start-up, RPM range, thrust and electrical limits on the bench.
Speed PI	Optional closed-loop RPM control	Add only when RPM is measurable and holding target RPM is a requirement.
Current limit / faults	Protection; not a PID loop	Measure current and voltage; define trip limits and safe response.
FOC current PI	Future i_d and i_q loops	Requires phase-current sensing and rotor electrical-angle estimation.
Shaft position loop	Not needed for a continuously spinning propeller	Only add if the motor must hold a specified shaft angle.

Aircraft position / track guidance is outside the Motor System. Label RPM and current as telemetry only where those measurements are actually available.